Bernoulli-EKF Multi-Target Tracker on SE(3)

A Minimal PMBM Reduction for Manipulation-Scene Object Tracking

1 Problem formulation

1.1 Setting and per-object state

A calibrated RGB-D camera observes a tabletop workspace populated by an unknown, time-varying number of rigid objects. A separate SLAM module is treated as a black box: it supplies, at every frame k, the camera pose $T_{cw,k} \in \mathrm{SE}(3)$ with covariance $\Sigma_{cw,k} \in \mathbb{R}^{6\times 6}$ in the $\mathfrak{se}(3)$ tangent. That is the only SLAM interaction in what follows.

The tracker maintains, online, a variable-size indexed family of *track parameters*

$$\mathcal{X}_k \;=\; \{X_k^{(i)}\}_{i\in\mathcal{I}_k}, \qquad X_k^{(i)} \;=\; \big(\,\ell^{(i)},\; T_k^{(i)},\; P_k^{(i)},\; r_k^{(i)}\,\big), \tag{1}$$

with four attributes per index: a class label $\ell^{(i)} \in \mathcal{L}$ (fixed at birth), a world-frame pose $T_k^{(i)} \in \mathrm{SE}(3)$, its covariance $P_k^{(i)} \in \mathbb{R}^{6\times 6}$ in the $\mathfrak{se}(3)$ tangent at $T_k^{(i)}$, and an existence probability $r_k^{(i)} \in [0,1]$ (the prior-probability that the object referred to by index i actually exists at time k). The index set $\mathcal{I}_k \subseteq \mathbb{N}$ is maintained by the tracker (birth enlarges it, pruning shrinks it).

The physical multi-target state at time k is a *random finite set* $\mathcal{T}_k \subseteq \mathrm{SE}(3)$ of world-frame poses, one per object actually present. Its (unknown) cardinality is $n_k = |\mathcal{T}_k|$. The parameter family $\mathcal{X}_k$ is the tracker's summary of the posterior on $\mathcal{T}_k$; the relationship is made precise in §4.

A per-object shape summary $\mathcal{G}_k^{(i)}$ (TSDF volume plus accumulated masked point cloud, rigidly attached to $T_k^{(i)}$) is maintained as a *by-product* of tracking: fused from depth after the pose is settled and $r_k^{(i)}$ is high, and fed back into the recursion only through a predicted-visibility predicate (§9). Geometric relations among objects and manipulation-role tags are deterministic predicates over $\mathcal{X}_k$, computed after each recursion step and consumed by downstream modules; they do not participate in the Bayesian update.

1.2 Per-frame measurements

Each frame delivers

$$Z_k \;=\; \big(\,I_k,\; D_k,\; \mathcal{D}_k\,\big), \qquad \mathcal{D}_k \;=\; \big\{\,(m_{k,l},\; \hat{\ell}_{k,l},\; \hat{s}_{k,l},\; \tau_{k,l})\,\big\}_{l=1}^{M_k}, \tag{2}$$

where I_k and D_k are RGB and depth images, and each entry of $\mathcal{D}_k$ is a binary mask $m_{k,l} \in \{0,1\}^{H\times W}$ with class label $\hat{\ell}_{k,l} \in \mathcal{L}$, calibrated detection score $\hat{s}_{k,l} \in [0,1]$, and a video-tracklet identifier $\tau_{k,l} \in \mathbb{N}$ issued by the segmenter (SAM2), which mostly preserves object identity across frames but is occasionally reset when the segmenter loses track – it enters the assignment cost (24) as a matching-ID bonus. All four fields are produced by the open-vocabulary detector + video-segmenter pipeline. **Detections live in pixel space; they do not carry 3D pose information.**

1.3 The mask-to-pose bottleneck

The feature distinguishing this setting from classical multi-target tracking is that *data association must be resolved before any pose measurement exists.* The segmenter tells us the image-space extent of each object hypothesis; converting that to a 3D observation requires a registration target, and that target is the tracker's own shape model $\mathcal{G}_{k-1}^{(i)}$. Each frame therefore executes

$$\mathcal{D}_k \xrightarrow{\text{back-projection}} \{\mathcal{C}_{k,l}\} \xrightarrow{\text{association}} \pi^* \xrightarrow{\text{ICP vs } \mathcal{G}_{k-1}^{(i)}} \{(\hat{T}_{k,l},\, \hat{R}_{k,l})\} \xrightarrow{\text{filter update}} \mathcal{X}_k. \tag{3}$$

Concretely, for each mask $m_{k,l}$:

1. **Back-projection.** The masked pixels of D_k yield a world-frame point cloud $\mathcal{C}_{k,l}$ via $T_{cw,k}$ and the intrinsics.
2. **Association.** A cost-matrix assignment (§6) decides whether $\mathcal{C}_{k,l}$ belongs to an existing track i, or spawns a new one.
3. **Pose measurement.** For a matched pair, colored ICP of $\mathcal{C}_{k,l}$ against $\mathcal{G}_{k-1}^{(i)}$, initialised at the prior mean $T_{k|k-1}^{(i)}$, produces a pose estimate $\hat{T}_{k,l} \in \mathrm{SE}(3)$ and a covariance $\hat{R}_{k,l} \in \mathbb{R}^{6\times 6}$ that folds ICP fitness/residual together with $\Sigma_{cw,k}$ through the SE(3) adjoint. For an unmatched mask we take $\hat{T}_{k,l}$ from the centroid of $\mathcal{C}_{k,l}$ with identity rotation and inflate $\hat{R}_{k,l}$ as a birth prior.

Only after (3) runs does the filter consume $(\hat{T}_{k,l}, \hat{R}_{k,l})$ as the measurement of an EKF update (§3). The tracker is therefore a feedback loop in which yesterday's state is what makes today's measurement well-defined.

1.4 Estimation target and pathologies

Conditioned on the SLAM pose stream, the Bayesian target is the multi-target set posterior

$$f_k(\mathcal{T}) \;\triangleq\; p\Big(\mathcal{T}_k = \mathcal{T} \;\Big|\; Z_{1:k},\, T_{cw,1:k},\, \Sigma_{cw,1:k}\Big). \tag{4}$$

The tracker represents f_k by the family $\mathcal{X}_k$ of (1) under the Multi-Bernoulli assumption of §4. Downstream we report the maximum-likelihood cardinality $\hat{n}_k$, per-track labels $\ell^{(i)}$, pose Gaussians $(T_k^{(i)}, P_k^{(i)})$, and existence probabilities $r_k^{(i)}$.

The recursion must absorb four failure modes simultaneously:

(i) *missed detections* – an object is present but yields no mask, either because the detector failed (probability $1 - p_d$) or because it is occluded / out-of-frustum (visibility p_v);

(ii) *clutter* – spurious masks not corresponding to any real object, at intensity λ_c;

(iii) *association ambiguity* – which mask belongs to which track; a wrong answer corrupts *both* the pose belief (wrong measurement) *and* the existence belief (spurious confirmation);

(iv) *model mismatch* – pose changes exceed what $P_{k|k-1}^{(i)}$ predicts, e.g. an object displaced by an unmodelled external agent, or ICP converging to a wrong local minimum.

Data association (iii) is one failure mode of four. The construction below treats it as a single pass through the same Mahalanobis-distance machinery that gates (ii) / (iv) and weights (i): a single scalar statistic d^2 drives every decision.

We formalise the recursion as a *Bernoulli-EKF tracker* on SE(3): per-object Gaussian pose beliefs with explicit existence probabilities, updated via a single-hypothesis Hungarian assignment

on the d^2 cost matrix, with birth from unmatched measurements and principled pruning from the existence update. This is the small-cardinality, low-clutter, single-hypothesis limit of the Poisson Multi-Bernoulli Mixture recursion [1, 2] – equivalently, the JIPDA reduction on SE(3).

2 SE(3) preliminaries

We treat poses as elements of the matrix Lie group SE(3) and represent small pose perturbations as 6-vectors $\xi = (\rho, \omega) \in \mathbb{R}^6$, collecting a translation $\rho \in \mathbb{R}^3$ and a rotation vector $\omega \in \mathbb{R}^3$. The mapping $\xi \mapsto \xi^\wedge \in \mathfrak{se}(3)$ to the Lie algebra $\mathfrak{se}(3)$ is the standard one [4]; its inverse is denoted $\xi = (\xi^\wedge)^\vee$. We invoke $\xi^\wedge$ only to state two basic operations in closed form, after which all computation happens directly in $\mathbb{R}^6$:

$$\mathrm{Exp} : \mathbb{R}^6 \to \mathrm{SE}(3), \qquad \mathrm{Exp}(\xi) \triangleq \exp(\xi^\wedge), \tag{5}$$

$$\mathrm{Log} : \mathrm{SE}(3) \to \mathbb{R}^6, \qquad \mathrm{Log}(T) \triangleq \log(T)^\vee. \tag{6}$$

Exp has a closed form (Rodrigues' formula in the rotation block, with a translation Jacobian) [4]; Log is its inverse. They are the only structural operations on SE(3) we use; group composition is plain matrix multiplication and inversion is the matrix inverse.

A *Gaussian on* SE(3) centred at $T_0 \in \mathrm{SE}(3)$ with covariance $P \in \mathbb{R}^{6\times 6}$ is the distribution of

$$T = T_0 \,\mathrm{Exp}(\xi), \qquad \xi \sim \mathcal{N}(\mathbf{0}, P), \tag{7}$$

written $T \sim \mathcal{N}(T_0, P)$. The mean lives in SE(3); the uncertainty lives in $\mathbb{R}^6$, attached to the tangent space at T_0.

3 Extended Kalman filter on SE(3) (single target)

For one rigid object, our process model is constant pose with $\mathbb{R}^6$ tangent-space process noise applied multiplicatively on the right:

$$T_k = T_{k-1} \,\mathrm{Exp}(\xi_k), \qquad \xi_k \sim \mathcal{N}(\mathbf{0}, Q_k). \tag{8}$$

Q_k encodes the per-frame dynamics prior (small when the object is stationary, larger during grasp/transport events; its value is selected externally by the manipulation context and is not estimated here). The measurement is an ICP-derived noisy pose (cf. §1.3), modelled the same way:

$$z_k = T_k \,\mathrm{Exp}(\eta_k), \qquad \eta_k \sim \mathcal{N}(\mathbf{0}, R_k). \tag{9}$$

R_k absorbs ICP uncertainty composed with the SLAM pose covariance $\Sigma_{cw,k}$.

Predict. Eq. (8) is *first-order identity* in the tangent at $T_{k-1|k-1}$: writing the prior as $T_{k-1|k-1}\,\mathrm{Exp}(\xi_{k-1|k-1})$ with $\xi_{k-1|k-1} \sim \mathcal{N}(\mathbf{0}, P_{k-1|k-1})$, the predicted pose is $T_k = T_{k-1|k-1}\,\mathrm{Exp}(\xi_{k-1|k-1})\,\mathrm{Exp}(\xi_k) = T_{k-1|k-1}\,\mathrm{Exp}(\xi_{k-1|k-1} + \xi_k + \frac{1}{2}[\xi_{k-1|k-1}, \xi_k] + \cdots)$ by Baker–Campbell–Hausdorff. Dropping second-order and higher brackets (the standard EKF-on-Lie-group linearisation) yields $\xi_{k|k-1} \approx \xi_{k-1|k-1} + \xi_k \sim \mathcal{N}(\mathbf{0}, P_{k-1|k-1} + Q_k)$, so the Jacobian of the process is $F_k = I_6$. Applying the *covariance-saturation projector*

$$\Phi : \mathbb{R}^{6\times 6} \to \mathbb{R}^{6\times 6}, \qquad \Phi(P) \triangleq \min\bigl(1,\, \mathrm{tr}\, P_{\max} / \mathrm{tr}\, P\bigr)\, P, \tag{10}$$

for a chosen design constant $P_{\max} \succ \mathbf{0}$ (a static uninformative prior, $\operatorname{tr} P_{\max} > 0$), the predict step is

$$T_{k|k-1} = T_{k-1|k-1}, \tag{11}$$

$$P_{k|k-1} = \Phi\big(P_{k-1|k-1} + Q_k\big). \tag{12}$$

Φ preserves symmetry and positive-definiteness (scaling by a positive scalar) and caps the trace at $\operatorname{tr} P_{\max}$: the recursion is self-bounded under arbitrarily long missing-observation streaks. Without it P grows linearly in the number of missed frames.

Update. The innovation, residual covariance, and Kalman gain are

$$\nu_k \;=\; \operatorname{Log}\big(T_{k|k-1}^{-1}\, z_k\big) \;\in\; \mathbb{R}^6, \qquad S_k \;=\; P_{k|k-1} + R_k, \qquad K_k \;=\; P_{k|k-1}\, S_k^{-1}. \tag{13}$$

The measurement Jacobian is $H_k = I_6$ to first order in ν_k: both z and T live in SE(3), and the Log is evaluated at the prior mean, so the linearisation is exact when $\nu_k = \mathbf{0}$ and degrades smoothly with $\|\nu_k\|$. The state correction is applied in the tangent and folded back via right-multiplication by Exp; covariance is updated in Joseph form for numerical stability:

$$T_{k|k} = T_{k|k-1}\, \operatorname{Exp}\big(K_k\, \nu_k\big), \tag{14}$$

$$P_{k|k} = \Phi\Big(\,(I_6 - K_k)P_{k|k-1}(I_6 - K_k)^\top + K_k R_k K_k^\top\Big). \tag{15}$$

The same projector Φ is applied after the update so a high-innovation observation cannot inflate $P_{k|k}$ above $P_{\max}$ either.

Innovation likelihood and Mahalanobis statistic. By (9) and the linearisation of Log at $T_{k|k-1}^{-1} T_k$, the innovation satisfies $\nu_k \sim \mathcal{N}(\mathbf{0}, S_k)$ under the null hypothesis $H_0^{(k)}$: "z_k was generated by the track whose prior is $\mathcal{N}(T_{k|k-1}, P_{k|k-1})$." The corresponding Gaussian density on $\nu_k \in \mathbb{R}^6$, which we use as the measurement likelihood for the Bernoulli update below, is

$$\mathcal{L}(z_k \,\big|\, T_{k|k-1}, P_{k|k-1}) \;\triangleq\; \mathcal{N}\big(\nu_k;\, \mathbf{0},\, S_k\big) \;=\; (2\pi)^{-3}|S_k|^{-1/2} \exp\big(-\tfrac{1}{2} d_k^2\big), \tag{16}$$

$$d_k^2 \;\triangleq\; \nu_k^\top S_k^{-1} \nu_k \;\sim\; \chi_6^2 \text{ under } H_0^{(k)}. \tag{17}$$

(16) is exact at $\nu_k = \mathbf{0}$ (where the Jacobian of $z \mapsto \operatorname{Log}(T_{k|k-1}^{-1} z)$ is I_6) and first-order accurate elsewhere; the tangent density agrees with the SE(3) density up to this Jacobian.

Robust update. Define an inner gate $G_{\text{in}} = \chi_6^2(0.95) \approx 12.59$ and an outer gate $G_{\text{out}} = \chi_6^2(0.9997) \approx 25$, and the re-weighting factor

$$w_k \;=\; \begin{cases} 1, & d_k^2 \le G_{\text{in}}, \\ \sqrt{G_{\text{in}}/d_k^2}, & G_{\text{in}} < d_k^2 \le G_{\text{out}}, \\ 0, & d_k^2 > G_{\text{out}}. \end{cases} \tag{18}$$

The update (13)–(15) is then modified as follows:

- If $w_k = 0$ (outer-gate reject): skip the update entirely, setting $T_{k|k} = T_{k|k-1}$, $P_{k|k} = P_{k|k-1}$; z_k is routed to the unassigned-track branch of the Bernoulli update (§7).
- Otherwise ($w_k > 0$): substitute $R_k \leftarrow R_k / w_k$ in (13)–(15). Note that K_k, S_k are recomputed with the inflated R_k/w_k; ν_k is unchanged.

(18) is Huber's redescending M-estimator on the Mahalanobis residual, applied to the re-weighted EKF.

4 Per-object Bernoulli state

To each index $i \in \mathcal{I}_k$ we associate a *Bernoulli component* with parameters $B^{(i)} = (r^{(i)}, \ell^{(i)}, T^{(i)}, P^{(i)})$, representing the single-target set posterior over a per-index random finite set $\mathcal{T}^{(i)} \subseteq \mathrm{SE}(3)$ with set density

$$f^{(i)}(\mathcal{T}) \;=\; \begin{cases} 1 - r^{(i)}, & \mathcal{T} = \varnothing, \\ r^{(i)}\,\mathcal{N}\big(T;\, T^{(i)}, P^{(i)}\big), & \mathcal{T} = \{T\},\ T \in \mathrm{SE}(3), \\ 0, & |\mathcal{T}| \geq 2. \end{cases} \tag{19}$$

Here $\mathcal{N}(\cdot; T^{(i)}, P^{(i)})$ is the Gaussian density on SE(3) of (7). In words: with probability $1 - r^{(i)}$ the i-th track does not exist (empty set); with probability $r^{(i)}$ it exists at exactly one pose $T \in \mathrm{SE}(3)$ distributed according to $\mathcal{N}(\cdot; T^{(i)}, P^{(i)})$. Classical Bernoulli-RFS normalisation $\int f^{(i)}(\mathcal{T})\,\delta\mathcal{T} = 1$ holds by construction (FISST set integral; see [1]). The pair $(T^{(i)}, P^{(i)})$ is the EKF state of §3; $r^{(i)}$ is the new variable PMBM contributes; $\ell^{(i)}$ is fixed at birth and does not enter the RFS marginalisation.

Multi-Bernoulli approximation. We assume the per-index Bernoulli sets $\mathcal{T}^{(1)}, \ldots, \mathcal{T}^{(n)}$ are statistically independent. The physical pose RFS of §1 is then

$$\mathcal{T}_k \;=\; \bigcup_{i \in \mathcal{I}_k} \mathcal{T}_k^{(i)}, \tag{20}$$

and its multi-target set density factorises as the convolution (FISST union) of the per-index Bernoulli densities. Equivalently, for any measurable subset $\mathcal{T} \subseteq \mathrm{SE}(3)$,

$$f_k(\mathcal{T}) \;=\; \sum_{\substack{\{\mathcal{T}^{(i)}\}_{i \in \mathcal{I}_k}: \\ \bigcup_i \mathcal{T}^{(i)} = \mathcal{T}}} \prod_{i \in \mathcal{I}_k} f^{(i)}\big(\mathcal{T}^{(i)}\big), \tag{21}$$

which is the Multi-Bernoulli approximation to (4). Independence is the only approximation introduced relative to a full PMBM mixture; the class labels $\{\ell^{(i)}\}$ ride along as track-identity metadata.

5 Bernoulli prediction

For each $i \in \mathcal{I}_{k-1}$, given a constant survival probability $p_s \in (0, 1]$, prediction is the standard PMBM step (eq. (43) in [2], simplified to constant p_s):

$$\begin{aligned} (T_{k|k-1}^{(i)}, P_{k|k-1}^{(i)}) \quad & \text{via EKF predict (11)–(12) with } Q_k^{(i)}, \\ r_{k|k-1}^{(i)} &= p_s\, r_{k-1|k-1}^{(i)}. \end{aligned} \tag{22}$$

p_s is the (observer-independent) physical survival probability of the object between consecutive frames. For static manipulation scenes, where objects neither move nor vanish on their own, the natural value is $p_s = 1$, which collapses (22) to identity on r. Setting $p_s < 1$ would model spontaneous physical disappearance (rare in our domain). Crucially, p_s is *not* visibility-dependent: occlusion-induced "invisibility" does not affect physical survival; it only affects the observer's ability to detect, which enters the recursion through $\tilde{p}_d$ in the missed update (§7). $Q_k^{(i)}$ is supplied by the manipulation context and reused unchanged from the existing implementation.

Object birth is folded into data association (§8); we do not maintain a continuous Poisson birth intensity.

6 Data association

For every track–detection pair $(i,l)\in\mathcal{I}_{k-1}\times\{1,\dots,M_k\}$ compute

$$\nu_{il} = \mathrm{Log}\big(T_{k|k-1}^{(i)\,-1}\hat{T}_{k,l}\big), \quad S_{il} = P_{k|k-1}^{(i)} + \hat{R}_{k,l}, \quad d_{il}^2 = \nu_{il}^\top S_{il}^{-1}\nu_{il}, \tag{23}$$

i.e. (13)–(17) with $(z_k, R_k) \leftarrow (\hat{T}_{k,l}, \hat{R}_{k,l})$ and $(T_{k|k-1}, P_{k|k-1}) \leftarrow (T_{k|k-1}^{(i)}, P_{k|k-1}^{(i)})$. The assignment cost matrix $C\in(\mathbb{R}\cup\{+\infty\})^{|\mathcal{I}_{k-1}|\times M_k}$ combines the Mahalanobis distance with a class-label feasibility gate, a χ_6^2 outer gate, and – when the segmenter exposes a tracklet identifier $\tau_{k,l}$ – an additive bonus on matching IDs:

$$C_{il} \;=\; \begin{cases} d_{il}^2 \;-\; \alpha\,\mathbb{1}\big[\tau_{k,l} = \tau_{k-1}^{(i)}\big], & \hat{\ell}_{k,l} = \ell^{(i)} \text{ and } d_{il}^2 \le G_{\mathrm{out}}, \\ +\infty, & \text{otherwise.} \end{cases} \tag{24}$$

Here $\tau_{k-1}^{(i)}\in\mathbb{N}\cup\{\varnothing\}$ is the tracklet ID with which track i was last matched ($\varnothing$ if never matched, in which case the indicator is 0), and $\alpha\ge 0$ is a tunable bonus in χ_6^2 units ($\alpha=0$ drops the ID term; the paper default $\alpha=4.4$ breaks ties between two same-class tracks whose d^2 differ by a few units without overriding a clear geometric mismatch).

The MAP assignment is

$$\pi^* \;=\; \arg\min_{\pi} \sum_{i\in\mathcal{I}_{k-1}\,:\,\pi(i)>0} C_{i,\pi(i)}, \qquad \text{s.t. } \pi(i)=\pi(j)>0 \Rightarrow i=j, \tag{25}$$

with $\pi:\mathcal{I}_{k-1}\to\{0,1,\dots,M_k\}$ and $\pi(i)=0$ encoding "track i unassigned". (25) is solved with scipy's `linear_sum_assignment` (Jonker–Volgenant); any cost-$+\infty$ entry forces $\pi^*(i)=0$. After solving, set $\tau_k^{(i)} \leftarrow \tau_{k,l}$ on matched pairs and leave $\tau_k^{(i)} = \tau_{k-1}^{(i)}$ on misses (so a transient tracklet drop does not erase the memory); birth initialises $\tau_k^{(i_{\mathrm{new}})} = \tau_{k,l}$. Measurements with no positive-assignment track are available for birth (§8). (25) is the MAP collapse of the PMBM mixture over global associations (JIPDA reduction).

Note on circularity: d_{il}^2 depends on $(\hat{T}_{k,l}, \hat{R}_{k,l})$, which is itself produced by ICP against $\mathcal{G}_{k-1}^{(i)}$. The orchestrator breaks the loop by running ICP once per track on a coarsely-gated candidate set (masks whose world-frame centroid is within a fixed window of $T_{k|k-1}^{(i)}$); (24) then refines the selection using the resulting pose measurement. Masks failing the coarse gate for all tracks go to birth with a centroid-initialised $\hat{T}_{k,l}$.

7 Bernoulli measurement update

The updates below are direct Bayes applications conditioned on the MAP association π^*. Let $E^{(i)}$ denote the event "track i exists," with prior $\Pr[E^{(i)}] = r_{k|k-1}^{(i)}$.

Track i associated with $z_{k,l}$ ($\pi^*(i)=l$). Given the MAP assignment, $z_{k,l}$ is drawn either from the track's Gaussian-on-SE(3) (if $E^{(i)}$, detected with prob. p_d) or from clutter (if $\neg E^{(i)}$, density λ_c). By Bayes,

$$\begin{aligned}\Pr[E^{(i)} \,\big|\, z_{k,l}\sim\text{track } i] &= \frac{\Pr[z_{k,l}\,\big|\,E^{(i)}]\,\Pr[E^{(i)}]}{\Pr[z_{k,l}\,\big|\,E^{(i)}]\,\Pr[E^{(i)}] + \Pr[z_{k,l}\,\big|\,\neg E^{(i)}]\,\Pr[\neg E^{(i)}]} \\ &= \frac{p_d\,\mathcal{L}_{il}\,r_{k|k-1}^{(i)}}{p_d\,\mathcal{L}_{il}\,r_{k|k-1}^{(i)} + \lambda_c\big(1-r_{k|k-1}^{(i)}\big)},\end{aligned} \tag{26}$$

where $\mathcal{L}_{il}$ is (16) evaluated at (i, l). Hence

$$\boxed{r^{(i)}_{k|k} \;=\; \frac{p_d\, \mathcal{L}_{il}\, r^{(i)}_{k|k-1}}{p_d\, \mathcal{L}_{il}\, r^{(i)}_{k|k-1} + \lambda_c \left(1 - r^{(i)}_{k|k-1}\right)}.} \tag{27}$$

The conditional state posterior $p(T^{(i)} \mid E^{(i)}, z_{k,l})$ is the EKF update of (14)–(15) with measurement $(z_k, R_k) = (\hat{T}_{k,l}, \hat{R}_{k,l}/w_{il})$ and Huber weight w_{il} from (18); if $w_{il} = 0$ the pair is rerouted to the unassigned branch below.

Track i unassigned ($\pi^*(i) = 0$). Define the *state-dependent* detection probability $\tilde{p}^{(i)}_d = p_d \cdot p^{(i)}_v$, where $p^{(i)}_v$ is the geometric visibility predicate of §9 evaluated at $T^{(i)}_{k|k-1}$. Under the standard PMBM measurement model this is a state-dependent $p_d(\cdot)$ in PMBM's notation (eq. (2) in [2]); the visibility just makes the dependence explicit. Absence of a track-i measurement is consistent with existence only through the missed-detection mechanism (probability $1 - \tilde{p}^{(i)}_d$); non-existence yields the event trivially. By Bayes on the event $E^{(i)} =$ "track i exists",

$$\boxed{r^{(i)}_{k|k} \;=\; \frac{(1 - \tilde{p}^{(i)}_d)\, r^{(i)}_{k|k-1}}{(1 - \tilde{p}^{(i)}_d)\, r^{(i)}_{k|k-1} + \left(1 - r^{(i)}_{k|k-1}\right)},} \tag{28}$$

which is the standard Bernoulli-misdetection update with state-dependent p_d (cf. García-Fernández et al. eq. (31)). The state $(T^{(i)}, P^{(i)})$ inherits from the prediction since no new information arrives: $T^{(i)}_{k|k} = T^{(i)}_{k|k-1}$ and $P^{(i)}_{k|k} = P^{(i)}_{k|k-1}$.

When $p^{(i)}_v \to 0$ (geometric occlusion or out-of-FOV) we have $\tilde{p}^{(i)}_d \to 0$, yielding $r^{(i)}_{k|k} \to r^{(i)}_{k|k-1}$ in (28): *no penalty for not seeing the unseen.* Long-term existence loss for genuinely missing-but-expected tracks comes from accumulating (28) over frames in which $p^{(i)}_v$ is high.

8 Birth: unmatched measurements

Each measurement index l with $\pi^*(i) \neq l$ for every $i \in \mathcal{I}_{k-1}$ is a *candidate birth.* Under the simplest birth model, each mask either corresponds to a newly-appearing object (prior intensity λ_b, measured in units of expected births per frame) or to clutter (intensity λ_c). Treat the detector score $\hat{s}_{k,l} \in [0, 1]$ as a calibrated likelihood ratio between the two: the unnormalised likelihood of the mask under the "new object" hypothesis is proportional to $\hat{s}_{k,l}$, and under the "clutter" hypothesis proportional to 1 (independent of the score). Bayes gives the posterior existence probability of the new track,

$$r^{(i_{\text{new}})} \;=\; \frac{\lambda_b\, \hat{s}_{k,l}}{\lambda_b\, \hat{s}_{k,l} + \lambda_c}, \tag{29}$$

which reduces to $\hat{s}_{k,l}$ when $\lambda_b = \lambda_c$ and smoothly interpolates between a pure-score confidence ($\lambda_b \gg \lambda_c$) and a clutter-dominated fallback ($\lambda_c \gg \lambda_b$). The other parameters of the fresh Bernoulli are initialised from the measurement itself:

$$\ell^{(i_{\text{new}})} = \hat{\ell}_{k,l}, \quad T^{(i_{\text{new}})} = \hat{T}_{k,l}, \quad P^{(i_{\text{new}})} = \hat{R}_{k,l}. \tag{30}$$

Since there is no pose prior on a freshly-born object, the posterior covariance collapses to the measurement noise [3]: this is the Bayesian one-step update of an improper uniform prior by a Gaussian likelihood. The new index i_{new} is added to $\mathcal{I}_k$; λ_b is a calibration constant per environment.

A track is *reported* to downstream consumers (planning, shape fusion, scene-graph computation) iff $r^{(i)}_{k|k} \geq r_{\text{conf}}$ (e.g. 0.5); below this threshold the track is *tentative* – still predicted, associated, and updated, but masked from the report $\widehat{\mathcal{X}}_k$.

9 Visibility predicate p_v

We adopt a bounded, deterministic approximation of the true observer-dependent visibility. Let $K \in \mathbb{R}^{3\times 3}$ be the camera intrinsic matrix and $\Omega_{\text{img}} = [0, W] \times [0, H] \subset \mathbb{R}^2$ the image domain. Define the camera projection

$$\pi_K : \mathrm{SE}(3) \to \mathbb{R}^2 \cup \{\infty\}, \qquad \pi_K(T) \;=\; \begin{cases} (Kp)_{1:2}/(Kp)_3, & p = T_{1:3,4},\ (Kp)_3 > 0, \\ \infty, & \text{otherwise}, \end{cases} \tag{31}$$

i.e. the pinhole projection of the object-origin in the camera frame onto the image plane (with depth-positive gate). For tracks $i \neq j$, let $A^{(i)}_k \subseteq \Omega_{\text{img}}$ be the image-space axis-aligned bounding box of track i's shape summary $\mathcal{G}^{(i)}_{k-1}$ projected through $T^{-1}_{cw,k} T^{(i)}_{k|k-1}$, and $\bar{d}^{(i)}_k$ its mean projected depth. Define the occlusion coefficient

$$\rho_{ij,k} \;\triangleq\; \mathbb{1}\big[\bar{d}^{(j)}_k < \bar{d}^{(i)}_k\big] \, \frac{|A^{(j)}_k \cap A^{(i)}_k|}{|A^{(i)}_k|} \;\in\; [0, 1], \tag{32}$$

which is the fraction of i's projected footprint that is both covered by j in the image and *in front of* i along the viewing ray. Distant or non-overlapping j contribute $\rho_{ij,k} = 0$. Then

$$p^{(i)}_v \;=\; \mathbb{1}\big[\pi_K\big(T^{-1}_{cw,k}\, T^{(i)}_{k|k-1}\big) \in \Omega_{\text{img}}\big] \;\cdot \prod_{j \in \mathcal{I}_{k-1} \setminus \{i\}} \big(1 - \rho_{ij,k}\big). \tag{33}$$

$p^{(i)}_v = 0$ when the projected centroid leaves the image, and the product shrinks with each in-front occluder. (33) evaluates in $O(|\mathcal{I}_{k-1}|^2)$ on bounding boxes; rendering-based refinement (pixel-exact depth buffers) is a drop-in upgrade that does not change any other equation.

10 Pruning and output

After update, prune any track with $r^{(i)}_{k|k} < r_{\min}$ (e.g. 10^{-3}). A track that survives a long sequence of unobserved-but-expected frames sees its existence collapse via repeated application of (28), and is removed naturally without an ad-hoc M-of-N counter.

The output at frame k is the labelled MAP estimate

$$\widehat{\mathcal{X}}_k = \Big\{ \big(i,\ \ell^{(i)},\ T^{(i)}_{k|k},\ P^{(i)}_{k|k},\ r^{(i)}_{k|k}\big) \ :\ i \in \mathcal{I}_k,\ r^{(i)}_{k|k} \geq r_{\text{conf}} \Big\}. \tag{34}$$

11 Frame-by-frame algorithm

1. **Predict.** For each $i \in \mathcal{I}_{k-1}$: state via (11)–(12); existence via (22).

2. **Measure.** For every mask $m_{k,l}$ back-project to a world-frame point cloud $\mathcal{C}_{k,l}$. For each candidate track i passing a coarse gate, run colored ICP to obtain $(\hat{T}^{(i)}_{k,l}, \hat{R}^{(i)}_{k,l})$; compute d^2_{il} via (23) and C_{il} via (24).

3. **Associate.** Solve (25) for π^*.

4. **Update each track i.**
   - if $\pi^*(i) = l$ and $w_{il} > 0$: apply (27) and EKF (14)–(15) with R_{il}/w_{il}; set $\tau_k^{(i)} \leftarrow \tau_{k,l}$;
   - else: compute $p_v^{(i)}$ via (33) and apply (28) (leaving $\tau_k^{(i)} = \tau_{k-1}^{(i)}$).

5. **Birth.** For each unassigned mask: centroid-initialised $\hat{T}$ and inflated $\hat{R}$, then (29), (30), and $\tau_k^{(i_{\text{new}})} \leftarrow \tau_{k,l}$.

6. **Prune.** Drop i with $r_{k|k}^{(i)} < r_{\min}$.

7. **Emit.** Report $\widehat{\mathcal{X}}_k$.

12 Reduction to existing code

- Bernoulli state $(r^{(i)}, \ell^{(i)}, T^{(i)}, P^{(i)}, \tau^{(i)})$ replaces the current per-track entry; two extra scalar fields added to `rbpf_state.RBPFState`: existence r and last-matched SAM2 tracklet ID τ.

- Predict (22), (11)–(12) edits `rbpf_state.predict_objects`; the saturation projector Φ replaces the bucket-switching in `process_noise_for_phase`.

- Cost matrix (24), assignment (25), and per-pair update (27) + Huber-weighted EKF update replace the hard upstream-ID lookup in `orchestrator._fast_tier` with a soft, ID-bonused Mahalanobis assignment: SAM2 IDs still drive identity when consistent, but association degrades gracefully to pure geometry when SAM2 drops an ID.

- (28) replaces the integer `miss_streak`; pruning at $r_{\min}$ replaces the M-of-N deletion test.

- $p_v^{(i)}$ via (33) adds a new `visibility.py` module ($\sim$60 LOC, depends only on $T_{cw,k}$ and the per-object shape summaries).

Calibrated parameters. The recursion is parameterised by seven scalars,

$$p_s \in (0,1], \quad p_d \in (0,1], \quad \alpha \geq 0, \quad \lambda_c > 0, \quad \lambda_b > 0, \quad r_{\text{conf}} \in (0,1), \quad r_{\min} \in (0, r_{\text{conf}}),$$

one symmetric positive-definite matrix $P_{\max} \in \mathbb{R}^{6\times 6}$ (the covariance saturation cap of (10)), and two χ_6^2-quantiles $G_{\text{in}} = \chi_6^2(0.95) \approx 12.59$, $G_{\text{out}} = \chi_6^2(0.9997) \approx 25$. Defaults for static manipulation scenes: $p_s = 1$ (rigid objects do not spontaneously disappear), $p_d = 0.9$, $\alpha = 4.4$ (mild tie-break toward SAM2 ID continuity; calibrate on a held-out sequence), λ_c estimated from a few static frames as the average per-frame detection count divided by image area, $\lambda_b = \lambda_c$, $r_{\text{conf}} = 0.5$, $r_{\min} = 10^{-3}$, $P_{\max} = \text{diag}(\sigma_{\text{trans}}^2 \cdot \mathbf{1}_3, \sigma_{\text{rot}}^2 \cdot \mathbf{1}_3)$ with $\sigma_{\text{trans}} = 0.25\,\text{m}$, $\sigma_{\text{rot}} = \pi/4$. Setting $p_s < 1$ models a known per-frame physical-removal hazard (e.g. unstable platforms, transient occluding clutter) and is purely domain-specific; setting $\alpha = 0$ switches off the SAM2 ID prior entirely and recovers the pure-geometry cost (24).

13 Approximations relative to PMBM

The full PMBM posterior of [2] is

$$f_k(\mathcal{T}) = \sum_{Y \uplus W = \mathcal{T}} \underbrace{f^p(Y)}_{\text{PPP, undetected}} \quad \underbrace{f^{mbm}(W)}_{\text{MB mixture, detected}} \quad , \quad f^{mbm}(W) \propto \sum_j \sum_{X_1 \uplus \cdots \uplus X_n = W} \prod_i w_{j,i}\, f_{j,i}(X_i).$$

Our recursion is its small-cardinality, low-clutter, single-hypothesis limit. We list below every place where it deviates and label each deviation as either an *approximation* (lossy collapse of an exact PMBM construct), a *modelling choice* (different but internally Bayesian model), or a *linearisation* (first-order truncation).

A.1 Single-hypothesis collapse via Hungarian assignment *(approximation).* We retain only the MAP global hypothesis j^* from $\sum_j w_j(\cdot)$, found by linear assignment (25). This is the JIPDA reduction of PMBM ([2] Sec. IV-B): conditional on j^* being the true assignment, all subsequent updates are exact Bayesian. Equivalent to PMBM's mixture only when the MAP assignment is unambiguous.

A.2 Marginalised single-MB existence form *(equivalence at marginal level).* PMBM sets $r^+ = 1$ on a successful association and pushes the ambiguity into the hypothesis weight $w_{j,i}$. After collapse to the MAP, those separately-tracked quantities marginalise to our (27): $r^+ = p_d \mathcal{L}\, r / (p_d \mathcal{L}\, r + \lambda_c(1-r))$. The misdetection update (28) is algebraically identical to PMBM's eq. (31) (with state-dependent $\tilde{p}_d^{(i)}$).

A.3 No PPP for never-detected targets *(modelling choice).* We do not maintain the Poisson intensity $\mu(\cdot)$ that PMBM uses to model targets that exist but have not yet produced a measurement. Equivalent to assuming every object in the scene eventually produces at least one detection, which holds for our manipulation domain (objects are deliberately placed by a human or robot and not invisible at first appearance). Loses PMBM's spatial prior over unseen states.

A.4 Categorical birth from detection score *(modelling choice).* We replace PMBM's birth integral $e(z) = p_d \int p(z|y)\mu(y)dy$ with a per-mask Bayes decision over a binary alternative "new object vs. clutter," using the detector score $\hat{s}_{k,l}$ as the likelihood ratio and rate-constants λ_b, λ_c as the prior odds: (29). Internally Bayesian under this model; not spatially adaptive.

A.5 Multi-Bernoulli factorisation across indices *(modelling choice).* We assume the per-index Bernoullis are statistically independent. PMBM also makes this assumption *within* each global hypothesis j; we make it *globally* after the MAP collapse.

A.6 Constant survival probability with $p_s = 1$ default *(modelling choice).* We follow standard PMBM in keeping p_s observer-independent; for static manipulation scenes the natural value is $p_s = 1$, removing predict-step decay entirely. Visibility-dependent "apparent" survival is recovered automatically through the state-dependent $\tilde{p}_d^{(i)}$ in the missed update, which is admissible PMBM.

A.7 EKF on $\mathrm{SE}(3)$ with $H = I_6$ *(linearisation).* Process and measurement noise are right-trivialised in the body tangent; the state composition and the innovation use BCH truncated at first order. Exact at $\nu_k = \mathbf{0}$; smoothly degrading with $\|\nu_k\|$. Standard EKF-on-Lie-group practice.

A.8 Covariance saturation Φ *(modelling choice).* The cap on $\operatorname{tr} P$ at $\operatorname{tr} P_{\max}$ in (10) is a pragmatic stabiliser for arbitrarily long missing-observation streaks; it equals an additive Gaussian prior $P_{\max}$ that is not in standard PMBM but is consistent with a maximum-entropy regularisation.

What this buys, what this costs. Approximations A.1 and A.2 are exactly the JIPDA reduction of PMBM, known to be a tight approximation when MAP associations are unambiguous and lossy when crossing tracks induce hypothesis ambiguity. Modelling choices A.3 – A.6 specialise the standard PMBM model to our domain and are reversible: restoring the PPP, Murty K-best mixture, spatially-adaptive birth, or per-class p_s would re-recover full PMBM at proportional computational cost. Linearisations A.7 – A.8 are inherent to the EKF-on-Lie-group and covariance-bounded layer; they do not change the multi-target structure of the recursion.

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